Toward a Theory of “Empty Mathematics”: A Foundational Framework for Observer-Dependent Abstraction

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Abstract

This paper introduces a conceptual framework called *Empty Mathematics* (“Ku-Mathematics”), intended as a pre-foundational layer underlying category theory, model theory, and other modern abstractions. Empty Mathematics formalizes two key cognitive operations: (i) the *emptying* (“ku-ization”) of structures into observer-dependent coherent states, and (ii) the *coloring* (“shiki-ization”) of these states into determinate mathematical objects. We argue that this bidirectional process provides a more primitive and flexible substrate than traditional foundations, enabling a principled account of multi-universe reasoning, observer variability, and non-isomorphic translations. We further propose that this framework is required to model certain extreme mathematical theories, such as Inter-Universal Teichmüller Theory (IUT), which inherently rely on incompatible or non-translatable structures.

1 Introduction

Classical foundations of mathematics—set theory, type theory, and category theory—presuppose that mathematical structures exist in a single determinate universe equipped with stable logical rules. However, several advanced theories (e.g., anabelian geometry, relative motives, and IUT) expose phenomena that cannot be internalized in a single universe without destructive loss of structure.

We introduce *Empty Mathematics* as a meta-level framework that explicitly models (1) observer dependence, (2) multi-universe inconsistency, and (3) non-resolution of certain structural relations. This framework extends beyond category theory by incorporating non-translatable states and systematically handling the alternation between undetermined and determined mathematical states.

2 Coherent and Decoherent Mathematical States

We define two complementary modes:

- **Coherent (Empty) States**: mathematical entities remain in an indeterminate state; morphisms, objects, and even universes are not yet fixed. These states represent “pre-mathematical” abstraction.
- **Decoherent (Colored) States**: specific mathematical structures become fixed; categories, sets, elements, and morphisms acquire determinate identity.

Empty Mathematics studies the controlled alternation between these states. Category theory corresponds to a fully decoherent phase, where composition, identity, and naturality are strictly enforced. In contrast, coherent states allow incompatible identifications and alternative observer positions.

3 Observer-Dependent Universes and F-Fix Points

Let F denote an observer configuration ("F-fix point"). A coherent state is defined relative to F. Different observers may induce incompatible decoherent states even when starting from identical coherent seeds.

This formalizes the notion that mathematical truth, when abstract enough, becomes partially observer-relative.

4 Non-Isomorphic Translations

One of the motivations for Empty Mathematics is to give a principled account of "non-isomorphic translations"—transitions between mathematical universes where no natural functor or equivalence exists. Empty Mathematics explicitly allows "colorings" that cannot be compared within any category.

Structures such as those encountered in IUT—where certain identifications are intentionally destroyed—fit naturally into this framework.

5 Toward Applications

Potential applications include:

- a meta-logic for multi-universe mathematics,
- a foundation for future AI systems capable of observer-variable reasoning,
- a conceptual simplification of extreme theories such as IUT,
- a cognitive model explaining creativity in high-level mathematics.

6 Conclusion

Empty Mathematics proposes a pre-foundational abstraction layer that models both coherent and decoherent phases of mathematical reasoning. Future work includes formalizing observer shifts, defining coherent-to-decoherent operators, and exploring connections with homotopy type theory and higher category theory.